Jihwan Moon

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Summary

- Applied Scientist at INFOCZ Inc. and Ph.D. candidate (**coursework completed**) at Seoul National University's Vision and Learning Lab, advised by Prof. Gunhee Kim; work spans **data-centric AI research** and **large-scale production ML**, from problem formulation and modeling to deployed systems.
- Build the ML core of **GOS**, an enterprise cybersecurity product that processes **1.4B+ provenance-graph edges per day** of EDR telemetry and is being deployed to **10K+ enterprise hosts**, scaling toward **200K**; owned its inference and serving path for a **10×+ speedup**.
- Published data-centric AI research at top venues: online video–text data filtering (**CVPR 2025**, first author), sample selection under noisy labels (**NeurIPS 2024**), and decoupled greedy block-wise learning (**ICLR 2021**).
- My research aims to **make AI more accessible and affordable**: local (greedy block-wise) learning and decentralized learning for LLM pretraining and fine-tuning. I also focus on **AI for cybersecurity**: **provenance-graph threat detection** and how such detectors are evaluated, **AI agent security**, and **data leakage protection**.

Professional & Research Experience

INFOCZ Inc. Seoul, South Korea
Applied Scientist Aug 2026 – Present

- Lead research on graph neural network architectures and self-supervised learning for advanced persistent threat (APT) detection in **GOS**, an enterprise cybersecurity product for large-scale EDR telemetry that combines provenance-graph modeling, GNN-based anomaly detection, and LLM-assisted filtering and investigation.
- Contributed to the provenance-graph processing and ML pipeline behind GOS — security-log ingestion, graph construction, feature generation, and anomaly detection at **1.4B+ edges per day** — now being deployed to **10K+ enterprise hosts**, with a planned rollout to **200K hosts**.
- Helped raise Specificity @ Recall $= 1.0$ from **69.5% to 99.7%** (GNN only) and **99.98%** (GNN + LLM filter); the system surfaced **18 attack behaviors** missed by the incumbent security system, spanning Lateral Movement, Discovery, Reconnaissance, Persistence, Command & Control, and Defense Evasion.
- First-authored an arXiv preprint auditing evaluation in provenance-based intrusion detection, the problem setting behind GOS: audited **nine** released implementations, ran **four** detectors with their original code, isolated three measurement effects behind reported scores, and proposed **STRICT**, six checkable reporting requirements.
- Optimized model inference and serving for GOS by designing a producer–consumer pipeline for better CPU–GPU overlap and scheduling, achieving a speedup of over **10×**.

KC ML2 Seoul, South Korea
Research Resident Jun 2026 – Aug 2026

- Led research on local (greedy block-wise) learning for LLM pretraining, reducing the memory and interconnect requirements of large-model training.
- First-authored *Greedy Local Learning for Language Model Pretraining: Gaps and Objective Design*, accepted to the NeurIPS 2026 Workshop on Collaborative, Open, and Decentralized Training of Foundation Models (CODEC-FM).
- Co-authored *REViT-v2*, a hierarchical windowed roto-reflection equivariant ViT that scales group-equivariant ViTs to ImageNet, accepted to the NeurIPS 2026 Workshop on Symmetry and Geometry in Neural Representations (Extended Abstract Track).

LG AI Research Seoul, South Korea
Research Scientist Intern, Bio Intelligence Lab Sep 2025 – Mar 2026

- Researched token compression and pruning for Vision Transformers processing gigapixel-scale inputs, including whole-slide images (WSIs), to cut the compute cost of full-resolution inference.

Allen Institute for AI (AI2)
Research Collaborator Jun 2024 – Feb 2025

- Co-led online video–text data filtering research with AI2 researchers for efficient training on web-scale data streams; published at CVPR 2025.

Vision and Learning Lab, Seoul National University Seoul, South Korea
Graduate Research Assistant (Ph.D. coursework completed) Mar 2021 – Present

- Conducted and led research on multimodal data filtering, noisy-label sample selection, and improving multimodal LLMs with a multimodal knowledge base (NeurIPS 2024, CVPR 2025).

Undergraduate Research Intern Jul 2020 – Feb 2021

- Searched for optimal neural network block configurations for greedy block-wise (local) learning, matching or beating backpropagation with asynchronous block training; published at ICLR 2021.

Platfarm Inc. Seoul, South Korea
NLP / Deep Learning Engineer Jun 2019 – Jan 2020

- Improved a production sticker and emoji recommendation system using NLP and deep learning.
- Won **1st place** in the EmotionGIF 2020 challenge.

Publications

Where the Numbers Come From: Auditing Evaluation in Provenance-Based Intrusion Detection arXiv 2026

Jihwan Moon, Gunhee Kim, Myeongjang Pyeon

Greedy Local Learning for Language Model Pretraining: Gaps and Objective Design NeurIPS 2026 Workshop

Jihwan Moon, Sheir A. Zaheer, Jinmyoung Lee, Gunhee Kim, Chan Y. Park
Workshop on Collaborative, Open, and Decentralized Training of Foundation Models (CODEC-FM)

REViT-v2: Hierarchical Windowed Roto-reflection Equivariant ViT for Equivariant Feature Extraction NeurIPS 2026 Workshop

Sheir A. Zaheer, ***Jihwan Moon***, Chan Y. Park
Workshop on Symmetry and Geometry in Neural Representations (Extended Abstract Track)

Refined Annotations and Specialized Augmentation for Minor Epicardial Arteries in Coronary Angiography Scientific Reports 2026

Do-Hyun Kim, Sangwoo Moon, Daehyeon Choe, Je Yoon Shin, Su Jin Kim, Jae Jin Lee, Daehong Kang, ***Jihwan Moon***, Si-Hyuck Kang

ReSpec: Relevance and Specificity Grounded Online Filtering for Learning on Video-Text Data Streams CVPR 2025

Jihwan Moon*, Chris Dongjoo Kim*, Sangwoo Moon, Heeseung Yun, Sihaeng Lee, Aniruddha Kembhavi, Soonyoung Lee, Gunhee Kim, Sangho Lee, Christopher Clark

Sample Selection via Contrastive Fragmentation for Noisy Label Regression NeurIPS 2024

Chris Dongjoo Kim*, Sangwoo Moon*, ***Jihwan Moon***, Dongyeon Woo, Gunhee Kim

SEDONA: Search for Decoupled Neural Networks Toward Greedy Block-wise Learning ICLR 2021

Myeongjang Pyeon, ***Jihwan Moon***, Taeyoung Hahn, Gunhee Kim

** Equal contribution.*

Education

Seoul National University Seoul, South Korea
Integrated M.S. & Ph.D. program in Artificial Intelligence Mar 2021 – Present
Ph.D. coursework completed

- Advisor: Prof. Gunhee Kim; GPA: 4.12/4.3

Seoul National University Seoul, South Korea
B.S. in Economics Mar 2013 – Feb 2021

- Minor in Computer Science; GPA: 3.67/4.3

Skills & Languages

Research: Data-Centric AI (data selection & filtering, noisy labels), Multimodal Representation Learning, Self-Supervised Learning, Efficient Training (local / block-wise, decentralized), LLM Pretraining & Fine-tuning, AI for Cybersecurity (intrusion & threat detection, AI agent security, data leakage protection), Graph Neural Networks, Anomaly Detection

Programming: Python **Frameworks:** PyTorch, TensorFlow

Systems: Inference & serving optimization (CPU–GPU overlap, scheduling), large-scale graph data pipelines, distributed training

Languages: Korean (native), English (fluent)

Academic Service & Teaching

Reviewer

- Journal: IEEE TPAMI (2024, 2025)
- Conferences: CVPR, ECCV, NeurIPS (2026)

Teaching Assistant (Seoul National University)

- Probabilistic Graphical Model (Spring 2022, Spring 2025)
- Discrete Mathematics (Spring 2023)
- Theory and Lab of IoT, AI and Big Data (Fall 2022)